

PF Lab 12 Tasks (4)

Name: Hassan Ali Khan

Roll No: 25K-0516

Section: BCS-1K

Task 1

```
#include<stdio.h>
#include<stdlib.h>

int main()
{
    int *arr = (int *)malloc(5 * sizeof(int));
    for(int i=0;i<5;i++)
    {
        arr[i] = i+1;
    }
    printf("value of 6th element: %d", arr[5]);
    free(arr);
    return 0;
}
```

 C:\Users\student25.KHIFAST\Documents\1.exe

```
value of 6th element: 1163282770
-----
Process exited after 0.02971 seconds with return value 0
Press any key to continue . . .
```

Task 2

```
#include<stdio.h>
#include<stdlib.h>

int main()
{
    int *arr = (int *)malloc(5 * sizeof(int));
    for(int i=0;i<5;i++)
```

```
{  
  
    arr[i] = i+1;  
  
}  
  
free(arr);  
free(arr);  
arr=NULL;  
printf(" %d", arr);  
return 0;  
}
```

 C:\Users\student25.KHIFAST\Documents\2.exe

```
-----  
Process exited after 0.1946 seconds with return value 3221226356  
Press any key to continue . . .
```

Task 3

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
int main()
```

```
{
```

```
    int *arr = (int *)malloc(5 * sizeof(int));
```

```
    if(arr==NULL)
```

```
    {
```

```
        printf("Memory Allocation Failed\n");
```

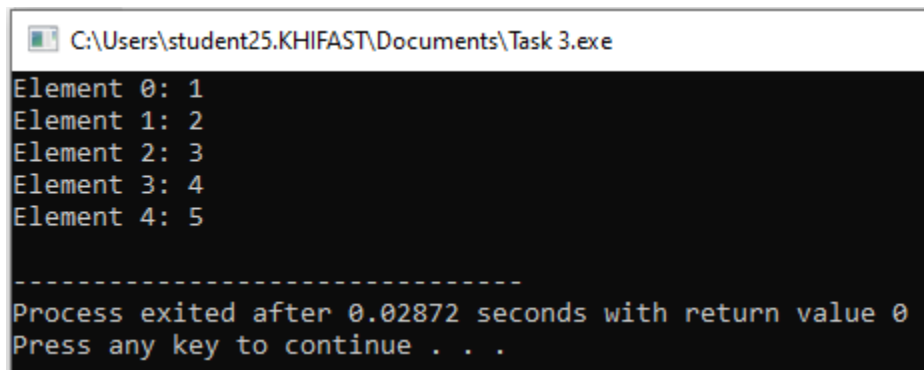
```
        return 1;
```

```
    }
```

```

    for(int i=0;i<5;i++)
    {
        *(arr+i)=i+1;
    }
    for(int i=0; i<5;i++)
    {
        printf("Element %d: %d\n",i, *(arr+i));
    }
    free(arr);
    return 0;
}

```



```

C:\Users\student25.KHIFAST\Documents\Task 3.exe
Element 0: 1
Element 1: 2
Element 2: 3
Element 3: 4
Element 4: 5
-----
Process exited after 0.02872 seconds with return value 0
Press any key to continue . . .

```

Task 4

```

#include <stdio.h>

#include <stdlib.h>

int main() {
    int *arr = (int *)malloc(2 * sizeof(int));
    if (arr == NULL) {
        printf("Memory Allocation Failed\n");
        return 1;
    }
}

```

```
arr[0] = 10;
```

```
arr[1] = 20;
```

```
int *temp = (int *)realloc(arr, 5 * sizeof(int));
```

```
if (temp == NULL) {
```

```
    printf("Memory Reallocation Failed\n");
```

```
    free(arr);
```

```
    return 1;
```

```
}
```

```
arr = temp;
```

```
printf("Array Resized to 5 elements\n");
```

```
for (int i = 2; i < 5; i++) {
```

```
    arr[i] = (i + 1) * 10;
```

```
}
```

```
for (int i = 0; i < 5; i++) {
```

```
    printf("Element %d: %d\n", i + 1, arr[i]);
```

```
}
```

```
free(arr);
```

```
return 0;
```

```
}
```

```
C:\Users\student25.KHIFAST\Documents\Task 4.exe
Array Resized to 5 elements
Element 1: 10
Element 2: 20
Element 3: 30
Element 4: 40
Element 5: 50

-----
Process exited after 0.02922 seconds with return value 0
Press any key to continue . . .
```

Task 5

```
#include <stdio.h>

#include <stdlib.h>

int main() {
    int rows=3,cols=3;
    int **p=(int**)calloc(rows, sizeof(int*));

    for (int i=0;i<rows; i++)
        p[i]=(int*)calloc(cols, sizeof(int));

    for (int i=0;i<rows;i++)
        for (int j = 0;j<cols;j++)
            scanf("%d",&p[i][j]);

    for (int i=0;i<rows;i++) {
        for (int j=0;j<cols;j++)
            printf("%d ",p[i][j]);
        printf("\n");
    }
}
```

```
return 0;
```

```
}
```

 C:\Users\student25.KHIFAST\Documents\Task 5.exe

